

Research Report: YOLO-Based Visual Recognition Model and STM32 Deployment Preparation

1. Project Overview

Objective: To train a lightweight model that detects chickens with "health" and "unhealth" labels, and separately explore model conversion and code generation with ST Edge AI for the STM32N6570-DK.

Tools Used: Python, Anaconda, YOLOv8 (Ultralytics), PyTorch, ONNX, and ST Edge AI.

Target Hardware: STM32N6570-DK Development Board [1].

2. What is YOLO?

YOLO (You Only Look Once): YOLO is a single-stage object detection algorithm. Compared with traditional two-step detection methods that first identify an object and then determine its category, YOLO performs object localization and classification in a single process, enabling faster detection.

Why YOLOv8n: The suffix "n" stands for Nano, which is the smallest model in the YOLOv8 family. Since the final deployment environment is a microcontroller, it is necessary to select a lightweight model with a small number of parameters, low computational requirements, and fast inference speed.

3. Implementation Process

I) Environment Setup

In a pure CPU environment using an Intel i5 processor, an independent Python virtual environment (yolo_env) was established using Anaconda to prevent dependency conflicts.

During the configuration of Ultralytics and PyTorch, errors were encountered due to a small number of images in .heic/.heif format. The configuration was modified to skip the abnormal file formats, ensuring that the training process could proceed smoothly.

II) Model Training

Data Preparation: More than 7,600 poultry images were processed. Before training, clear data classification criteria were established. Chickens were classified as "unhealthy" based on abnormal physical characteristics such as lying down, messy or dirty feathers, exposed skin, and twisted postures. Path mismatch errors between the cloud-based dataset paths and the local hard-drive paths in dataset.yaml were also identified and fixed.

Model Training: The model was trained using YOLOv8n pre-trained weights for 3 epochs on a CPU. The model achieved an mAP@50 of 81.9%.

III) Performance Validation and ONNX Export

The model was tested using new images. It successfully detected the chickens, drew bounding boxes, and output "health" and "unhealth" labels together with confidence scores. The confidence scores were within the range of 0 to 1.

As part of the YOLO experiment, the PyTorch .pt model was exported to ONNX. This export is distinct from the two-class ONNX model currently archived with the ST-generated files.

4. Summary

This project combined a YOLOv8n chicken-detection experiment with a separate ST Edge AI model-conversion exercise.

The archived my_model_OE_3_3_1.onnx and its ST-generated stai_network.c/h files describe a two-class image classifier with a 96×96 image input and two class scores as output. They are not the YOLOv8n chicken detector shown in the sample images. The class names and exact provenance of this archived classifier remain unconfirmed.

The ST conversion exercise also reached zero-error cross-compilation in STM32CubeIDE. With no STM32N657 hardware available, I did not perform flashing or on-board tests. Complete application integration, NPU execution, latency and memory use remain unverified.

5. Additional Information

Planned Deployment Process:

1. Confirm the chicken-detection model's compatibility with ST Edge AI, generate its C code, and verify supported operators and NPU mapping.
2. Use STM32CubeIDE to develop the firmware so that the STM32N6570-DK development board can automatically complete the entire process of capturing an image, sending it to the AI model for recognition, and outputting the result.
3. Flash the compiled program onto the microcontroller and test the actual operating speed and memory consumption under real-world conditions to verify that the software and hardware can work together properly.

[1] About the STM32N657 Development Board

The STM32N657 is a new generation of microcontroller featuring an integrated NPU. Its main processor is an ARM Cortex-M55, which operates at up to 800 MHz and supports Helium vector acceleration.

Compared with a conventional multi-core PC, it does not have the same level of general-purpose computing performance. However, it is specifically designed for edge computing and can efficiently process specific AI tasks with very low power consumption.

Its key hardware features include:

Independent Computing Power: The integrated NPU provides up to 600 GOPS of theoretical computing performance for neural network acceleration. This project has not measured on-board inference performance.

Memory and Camera Input: The device provides 4.2 MB of contiguous embedded SRAM and a CSI-2 camera interface for image input. These features support edge-vision development; end-to-end latency must be measured on the completed application.

Hardware Offloading: The integrated ISP handles image processing tasks such as image noise reduction, further reducing the workload on the CPU.

The platform also features a NeoChrom 2.5D GPU and a 5-inch touchscreen display with an 800×480 resolution, providing responsive user interface capabilities.

Overall, the STM32N657 is well suited to this project because it combines low-power MCU operation with dedicated AI acceleration for edge deployment.

6. References

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Author's Declaration: The conceptual framework and dataset were provided by the supervising professor. The software engineering, model training, debugging, and ONNX conversion were carried out by the author and owner of this GitHub repository, with reference to relevant technical documentation and development resources.

Chinese Version (original)

基于 YOLO 的视觉识别模型与 STM32 的部署准备报告

1、项目概述

- 目标: 训练一个轻量级模型, 对鸡进行目标检测并标注 health/unhealth 类别; 另行探索面向 STM32N6570-DK 的 ST Edge AI 模型转换与代码生成流程。
- 用到的工具: Python, Anaconda, YOLOv8 (Ultralytics), PyTorch, ONNX, ST Edge AI。
- 参考硬件: STM32N6570-DK 开发板 [\[1\]](#)。

2、YOLO 是什么?

- YOLO (You only look once): 一种单阶段目标检测算法。对比传统算法“先识别目标, 再判断类别”的两步识别方法, YOLO 将图片划分网格, 一次性输出所有的物体位置和类别, 速度更快。
- 为什么选 YOLOv8n: 后缀的 n 代表 Nano (最小号模型) 这是因为我们最终的运行环境是单片机, 所以必须选择参数量最少, 计算量最小, 推理速度最快的轻量级模型。

3、具体流程

I) 环境搭建

- 在纯 CPU 环境下 (Intel i5), 使用 Anaconda 搭建了独立的 Python 虚拟环境 (yolo_env), 防止依赖冲突。
- 配置 Ultralytics 和 PyTorch 时, 遇到少量 .heic/.heif 格式的图片导致安装报错。通过修改配置跳过异常文件, 确保了训练流程顺利推进。

II) 模型训练

- 数据准备: 处理 7600+ 张禽类图片, 图片在训练前明确了数据分类标准 (如基于禽类躺卧、毛发凌乱肮脏、皮肤裸露、姿态扭曲等异常特征来定义“不健康”), 修复了 dataset.yaml 中云端路径与本地硬盘不匹配的报错。
- 模型训练: 基于 YOLOv8n 预训练权重, 在 CPU 上训练了 3 个 epoch。此次初步实验记录的目标检测指标 mAP@50 为 81.9%。

III) 效果验证与 ONNX 导出

- 用新图片测试, 模型能成功识别并画出边框, 输出 health/unhealth 标签和置信度在 0 和 1 之间。
- 在 YOLO 实验中, 将 PyTorch 的 .pt 模型导出为 ONNX 格式。这项导出工作与当前随 ST 生成文件归档的二分类 ONNX 模型应分别说明。

4、总结

本项目包含 YOLOv8n 鸡目标检测实验, 以及另行开展的 ST Edge AI 模型转换练习。当前归档的 my_model_OE_3_3_1.onnx 及对应的 stai_network.c/h 文件描述的是输入为 96×96 图像、输出为两个类别分数的图像二分类模型, 并非示例图片所展示的 YOLOv8n 鸡目标检测模型; 该二分类模型的类别名称和具体来源尚未确认。ST 转换练习完成了代码生成, 并在 STM32CubeIDE 中实现了零错误交叉编译。由于没有 STM32N657 硬件, 我未进行烧录或上板测试; 完整应用集成、NPU 执行、运行延迟和内存占用仍有待验证。

5、补充

后续计划部署流程:

- 确认鸡目标检测模型与 ST Edge AI 的兼容性, 生成相应 C 代码, 并核查算子支持及 NPU 映射情况。
- 用 STM32CubeIDE 写代码, 让 STM32N6570-DK 开发板能自动完成“拍照、传给 AI 识别、输出结果”这一整套动作。
- 把编译好的程序烧录到单片机里, 测一下真实环境下的运行速度和内存占用情况, 确认软件和硬件能完美配合。

[1]: 关于 STM32N657 开发板

这是一块内置 NPU 的新一代微控制器。主控 CPU 为 ARM Cortex-M55 (单核 800 MHz, 支持 Helium 矢量运算加速)。相比传统多核 PC, 它虽在通用算力上不具优势, 但专为边缘端设计, 能以极低功耗高效处理特定 AI 任务。其核心硬件特性包括:

- 独立算力:内置 NPU 提供最高 600 GOPS 的理论算力, 专注神经网络加速;本项目尚未进行板端推理性能测试。
- 存储与图像输入:芯片提供 4.2 MB 连续片内 SRAM, 并配有 CSI-2 摄像头接口。这些能力为边缘视觉应用提供硬件基础, 实际端到端延迟仍需完整应用上测试。
- 硬件减负:内置 ISP 处理图像降噪, 进一步释放 CPU 算力;搭载 NeoChrom 2.5D GPU 及 5 英寸触控屏(800*480), 提供低延迟的 UI 响应。综上, 该硬件通过精简通用算力换取 AI 专精效能, 非常契合本项目的端侧部署需求。

6、参考资料

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